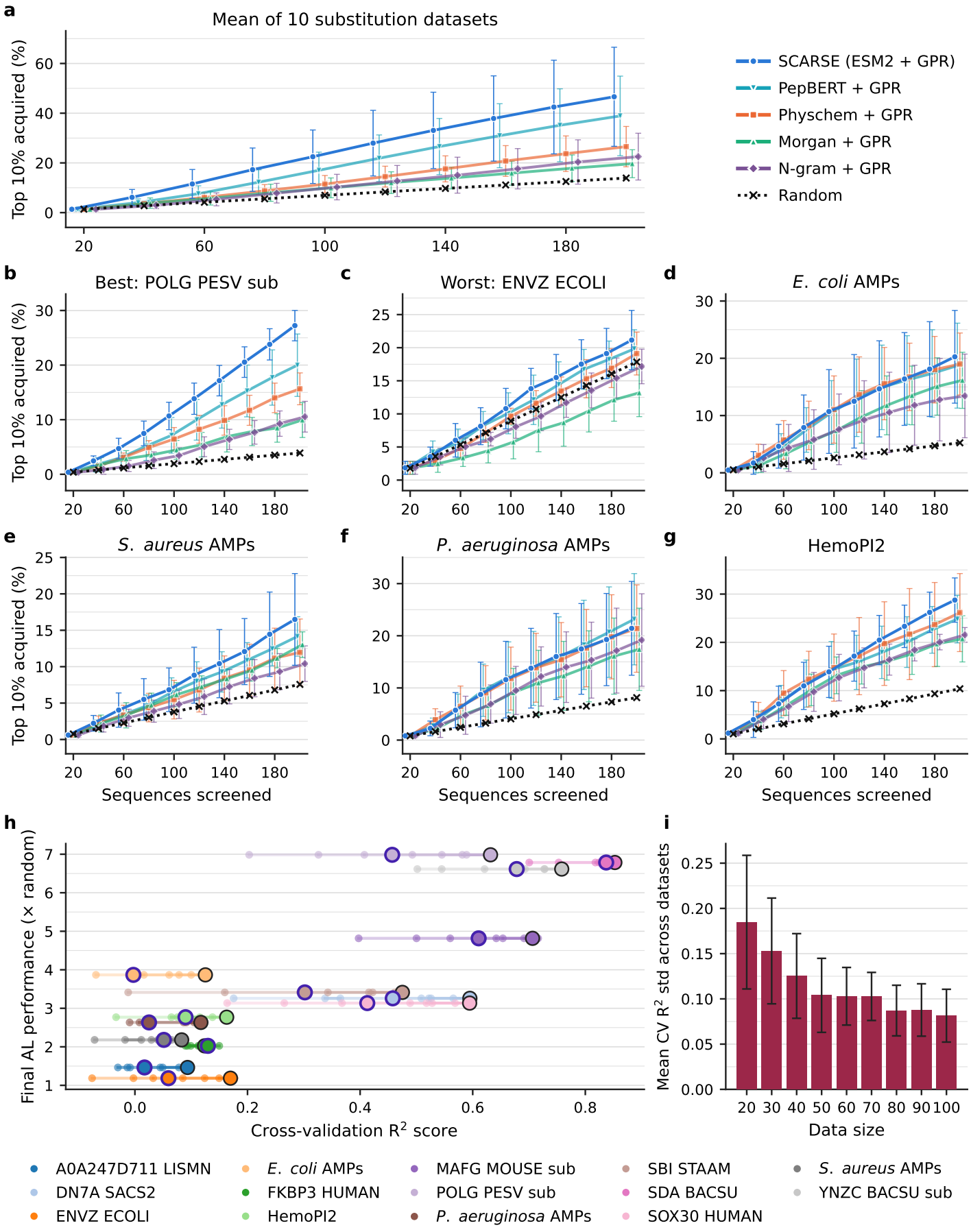

a
Mean of 10 substitution datasets
Top 10% acquired (%)
SCARSE (ESM2 + GPR)
PepBERT + GPR
Physchem + GPR
Morgan + GPR
N-gram + GPR
Random
b
Best: POLG PESV sub
c
Worst: ENVZ ECOLI
d
E. coli AMPs
e
S. aureus AMPs
f
P. aeruginosa AMPs
g
HemoPI2
Sequences screened
h
Final AL performance (× random)
Cross-validation R² score
i
Mean CV R² std across datasets
Data size
A0A247D711 LISMN
DN7A SACS2
ENVZ ECOLI
E. coli AMPs
FKBP3 HUMAN
HemoPI2
MAFG MOUSE sub
POLG PESV sub
P. aeruginosa AMPs
SBI STAAM
SDA BACSU
SOX30 HUMAN
S. aureus AMPs
YNZC BACSU sub